

# Computer-aided drug design (CADD)

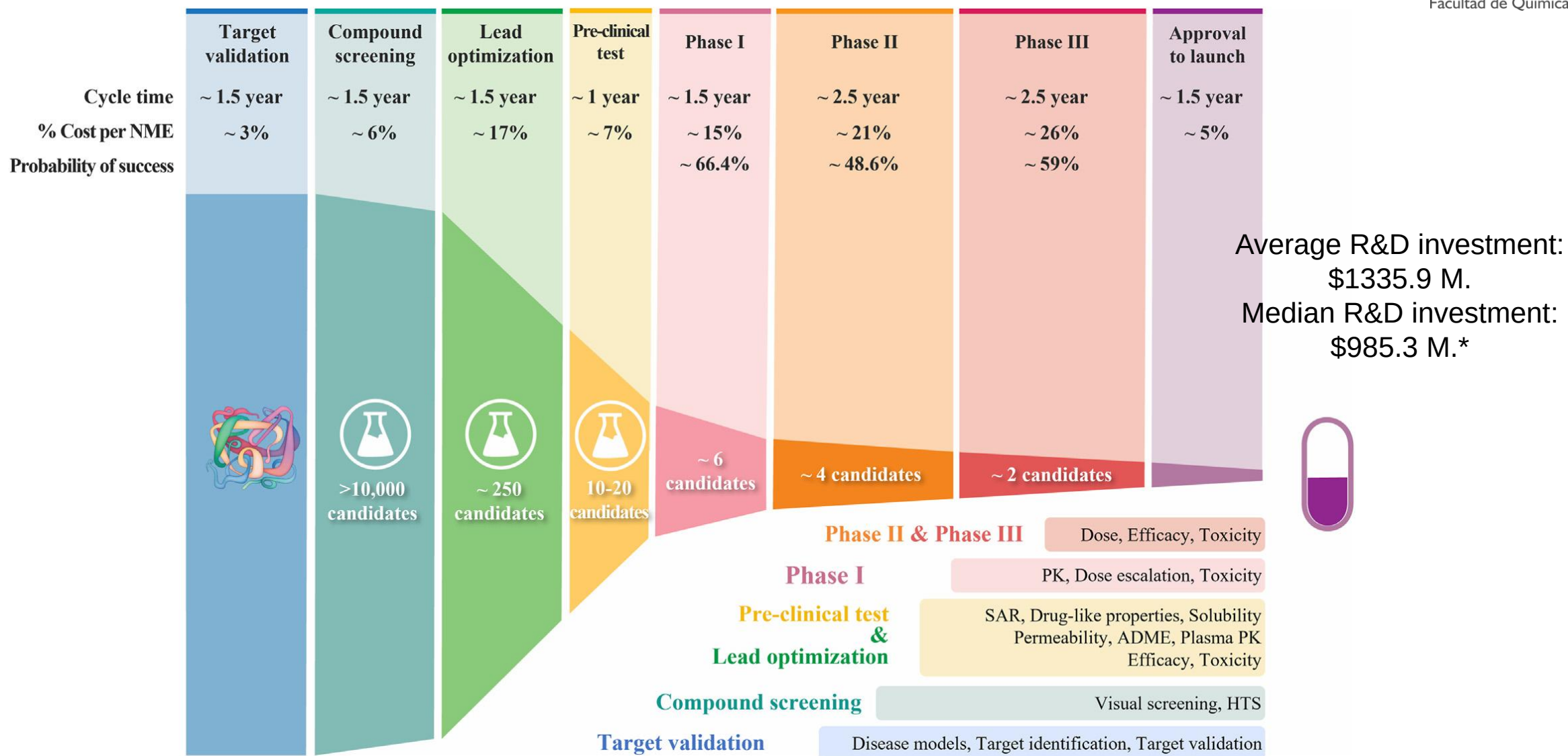
Juan David Sánchez Ramírez

Professor  
Dr. José Luis Medina Franco



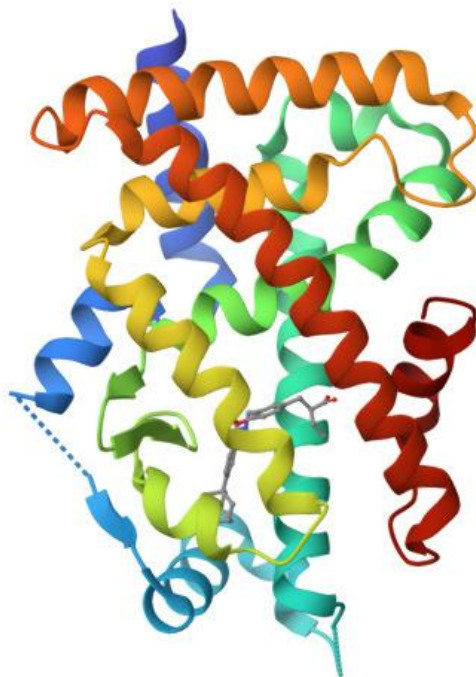
Facultad de Química  
Universidad Nacional Autónoma de México

# From Traditional to CADD-driven pipelines



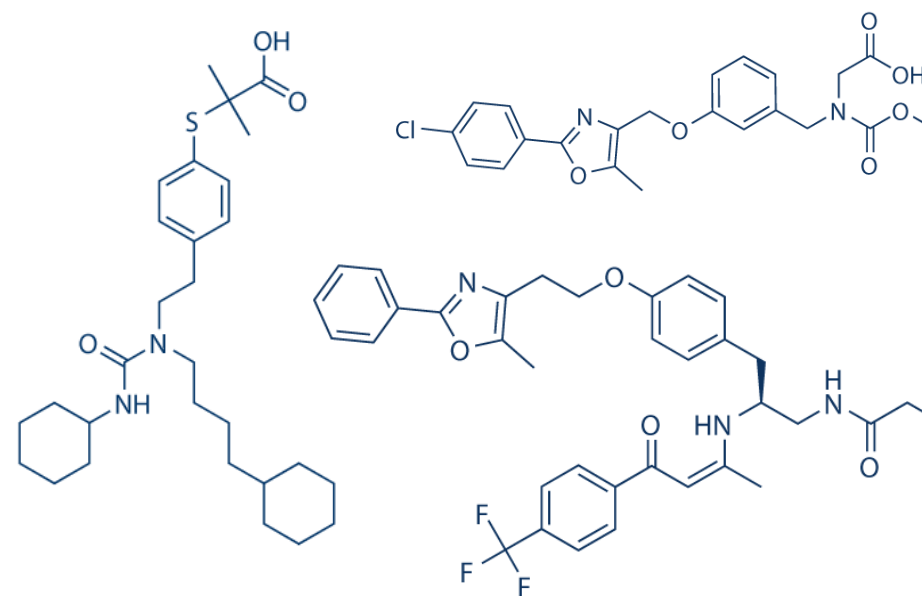
# Computer-aided drug design (CADD) methodologies

## Structure-based drug design (SBDD)



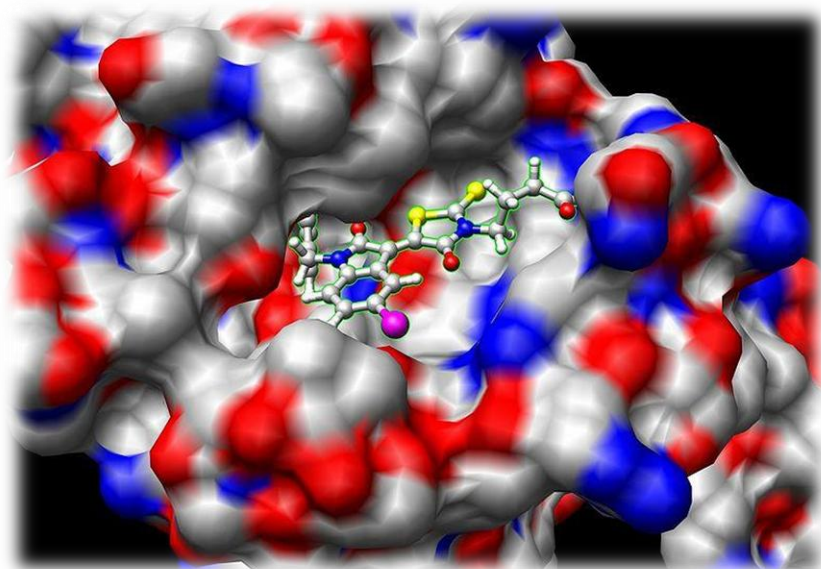
PPAR- $\alpha$

## Ligand-based drug design (LBDD)

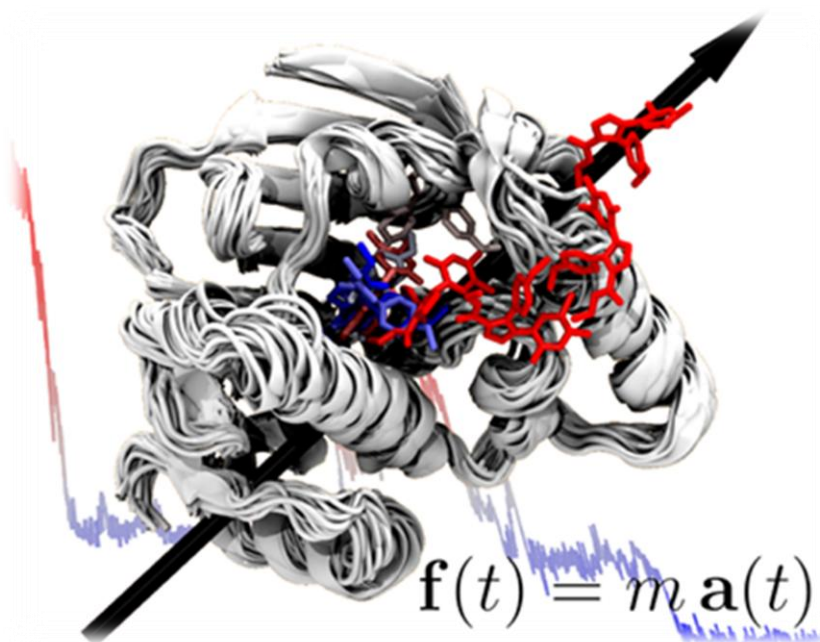


PPAR- $\alpha$  inhibitors

# Structure-based drug design (SBDD)



Molecular docking

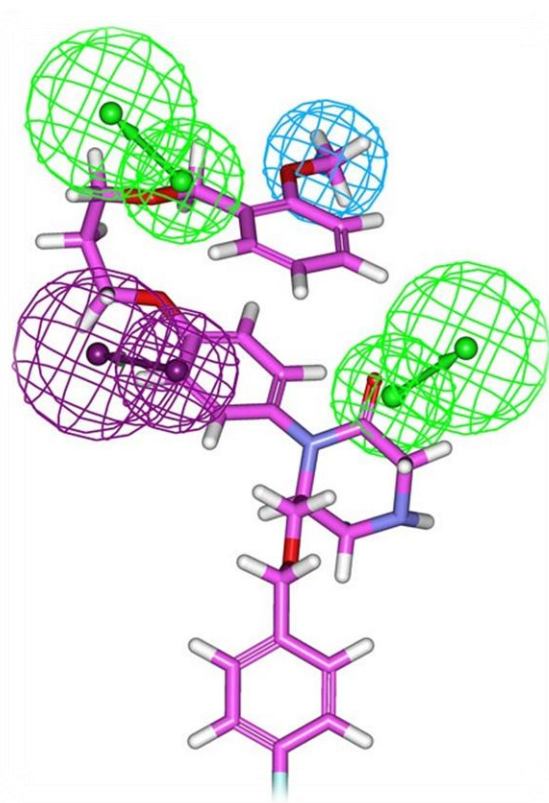


Molecular dynamics

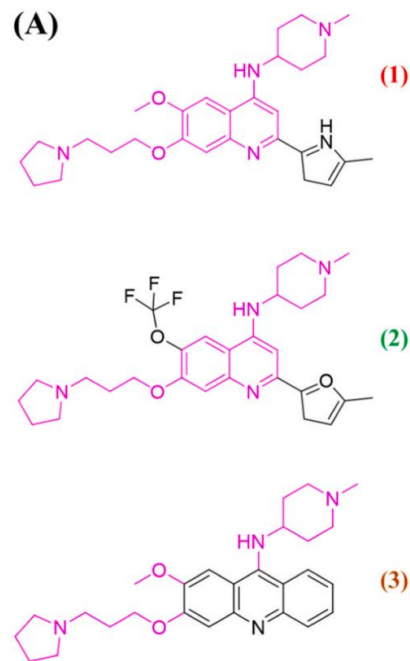
## Software:

AutoDock, Glide,  
Gold, Vina, MOE.  
GROMACS, AMBER.

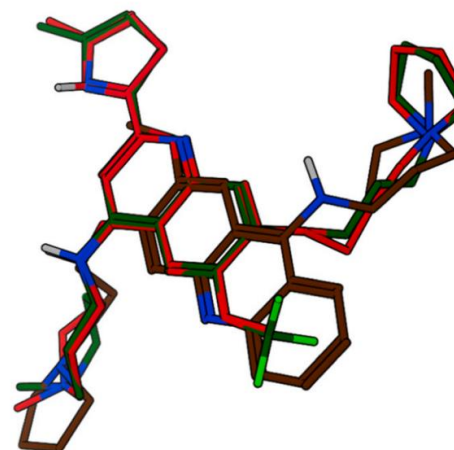
# Ligand-based drug design (LBDD)



Pharmacophore modeling



Molecular similarity



## Software:

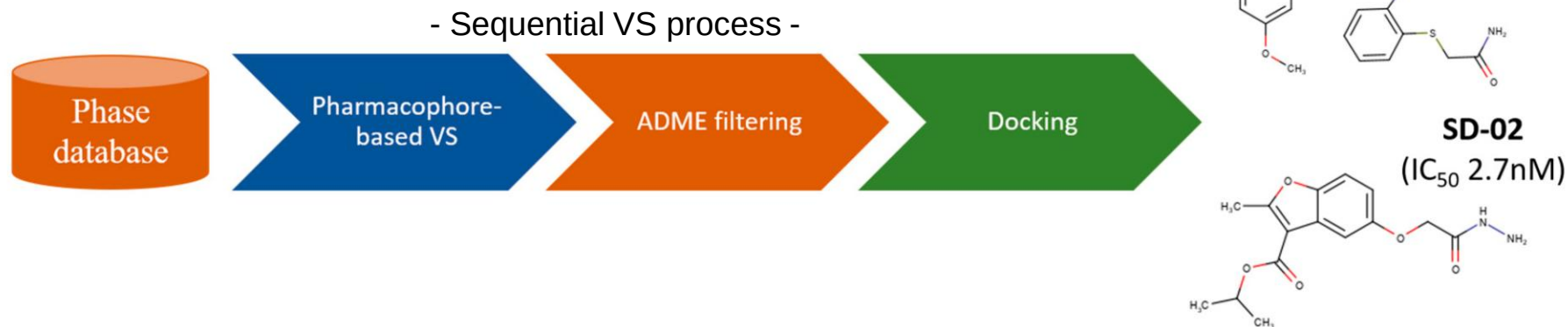
Schrödinger Phase,  
Pharmer, MOE.  
RDKit, SwissSimilarity.

# Hybrid approaches

**A.**



**B.**

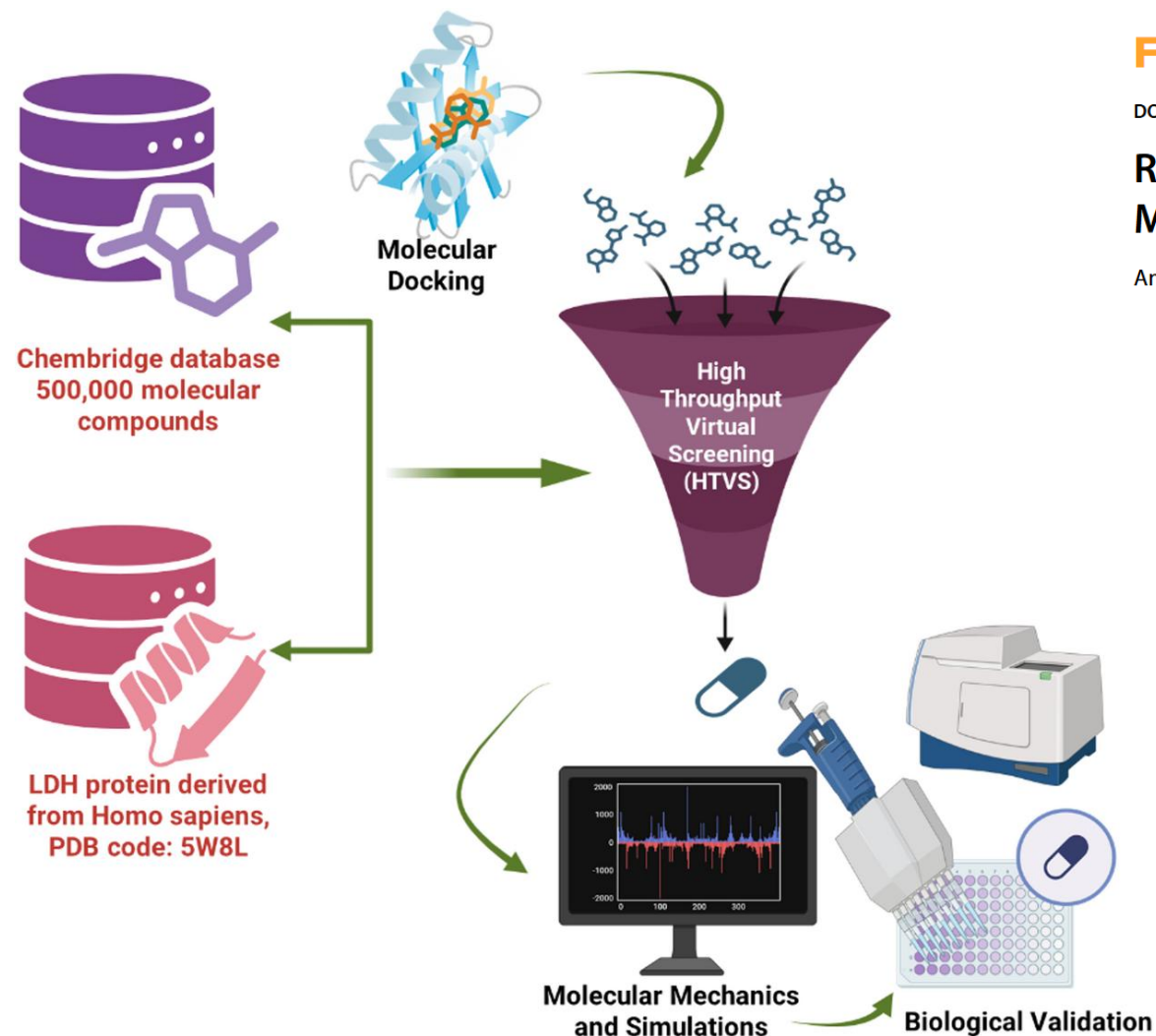


 SBDD Method

 LBDD Method

Representative cases of two combined ligand-based (LB) and structure-based (SB) strategies leading to the discovery of potent inhibitors of (A) 17 $\beta$ -hydroxysteroid dehydrogenase type 1 (17 $\beta$ -HSD1) and (B) histone deacetylase 8 (HDAC8) enzymes.

# High throughput virtual screening (HTVS)



## Full Paper

[www.molinf.com](http://www.molinf.com)

DOI: 10.1002/minf.202000028

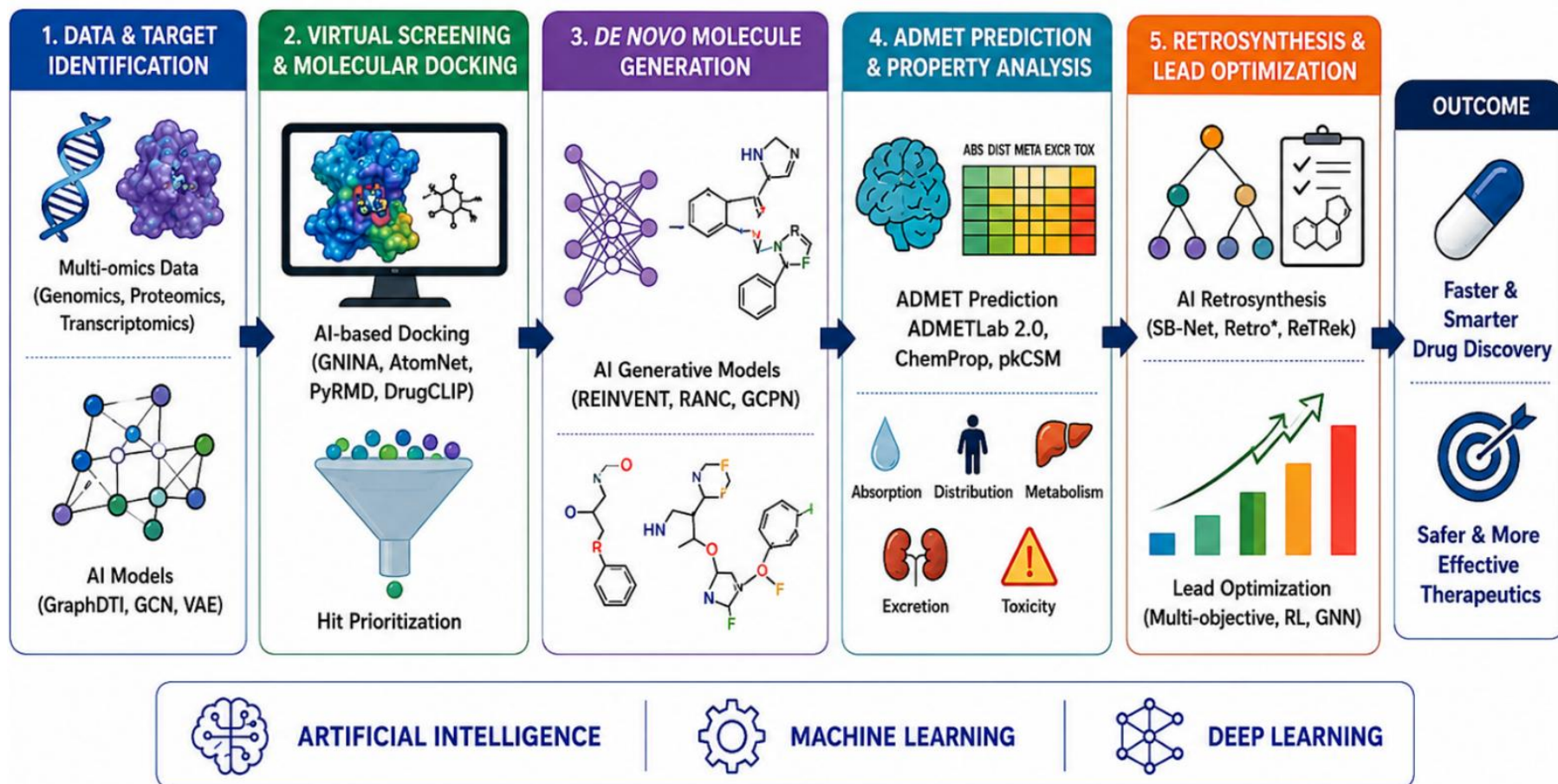
## Rapid Identification of Potential Inhibitors of SARS-CoV-2 Main Protease by Deep Docking of 1.3 Billion Compounds

Anh-Tien Ton<sup>+, [a]</sup> Francesco Gentile<sup>+, [a]</sup> Michael Hsing<sup>+, [a]</sup> Fuqiang Ban<sup>+, [a]</sup> and Artem Cherkasov<sup>\*, [a]</sup>

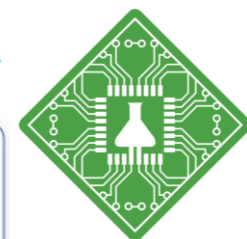
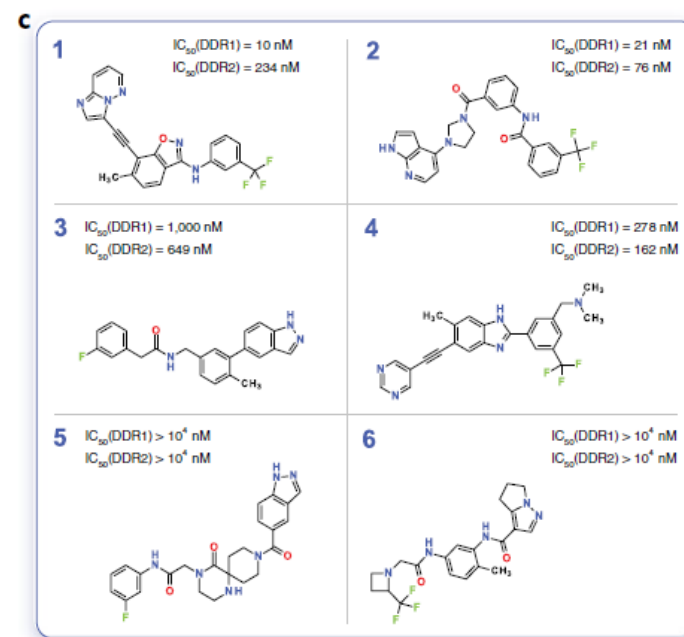
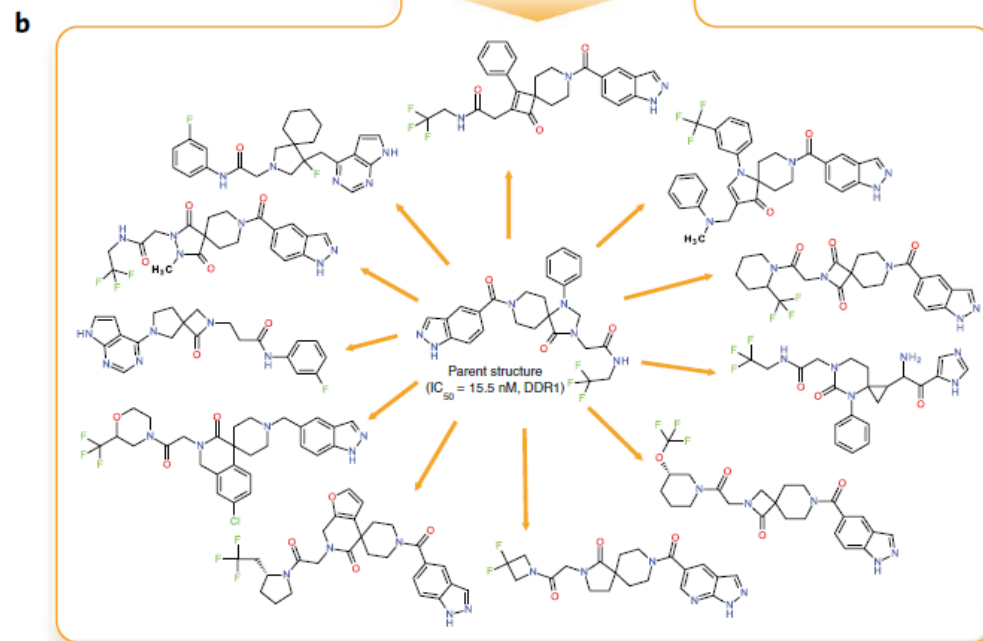
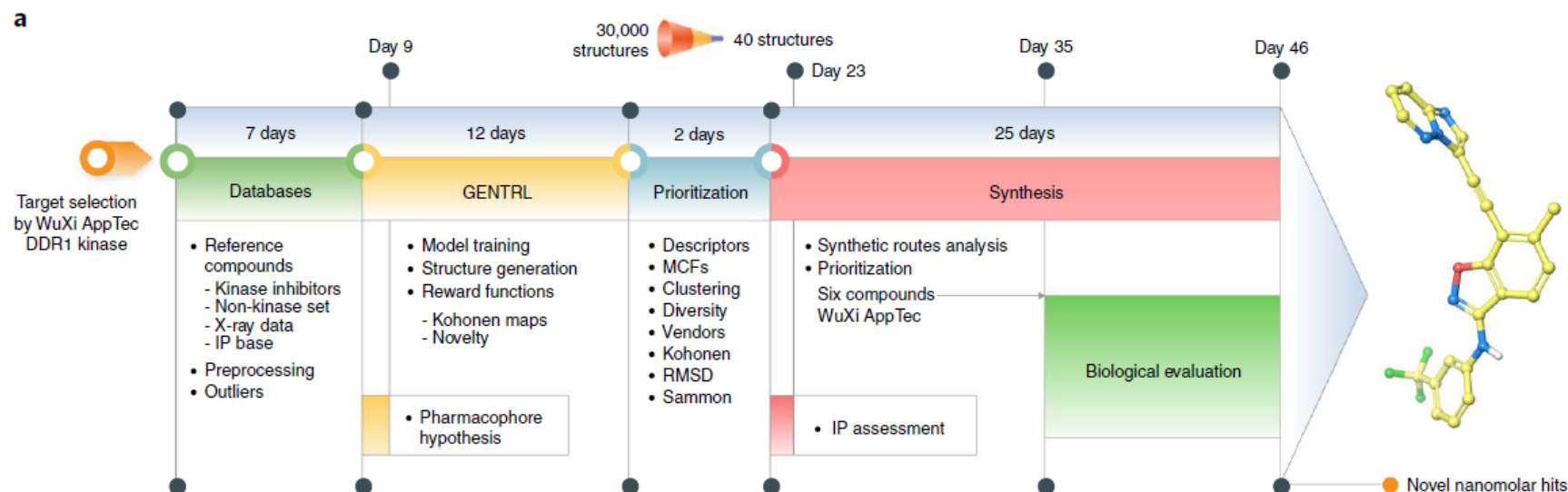


## AI-DRIVEN DRUG DISCOVERY

*Tools, Integrated Workflow, and Current Challenges*



# Frontiers in CADD



**Insilico  
Medicine**

# Frontiers in CADD

Generative AI Software for Drug Discovery, Scientific Research & Sustainability

**SOFTWARE FOR SCIENCE**

**Insilico Medicine**  
insilico.com

**THERAPEUTIC PIPELINE**

High Quality Therapeutic Programs Discovered Using Generative AI and Automation

**Biology<sup>42</sup>**

**PandaOmics**

Discover and prioritize novel targets

**Generative Biologics**

Discover and optimize novel biomolecules

**Life Star 2**

Automated lab operating environment

**Large Language of Life Models (LLMs)**

**Precious1GPT**

Multimomics age prediction & target ID

**Precious2GPT**

Multimodal multimomics biological data synthesis

**Precious3GPT**

Multi tissue multispecies multimomics multimodal life model

**Chemistry<sup>42</sup>**

**Generative Chemistry**

Generate novel molecules

**Retrosynthesis**

Predict synthetic routes for molecular structures

**Model Training**

Custom models based on in-house data

**MDFlow**

End-to-end simulation workflows for biomolecules and complexes

**Alchemy**

Physics-based relative binding free energy engine

**ADMET & Off-target**

Profiling & optimization

**MolSpace**

Visualize and analyze chemical data using dimensionality reduction technique

**Nach01**

Multimodal natural & chemical languages foundation model

**Medicine<sup>42</sup>**

**inClinico**

Design and predict clinical trials

**Science<sup>42</sup>**

**DORA**

Multi-agent generative research assistant

**LLM Assistant**

**Copilot**

Generative conversational agent

**Data Warehouse**

Seamless cross-application data flow via efficient integration & standardization

**Environmental Sustainability**

Generative AI technologies for environmental sustainability

**Science MMAI Gym**

Boost the LLM's intelligence in drug discovery and development

INDICATION

|                                  | TARGET ID | HIT TO LEAD           | LEAD OPT. | IND-ENABLING | PHASE I     | PHASE II   | PHASE III                                                |
|----------------------------------|-----------|-----------------------|-----------|--------------|-------------|------------|----------------------------------------------------------|
| Idiopathic Pulmonary Fibrosis    | TNIK      |                       |           |              | China       | China NMPA |                                                          |
| Idiopathic Pulmonary Fibrosis    | TNIK      |                       |           |              | New Zealand | US FDA     |                                                          |
| IPF Inhalable                    | TNIK      |                       |           |              |             |            | IND clearance                                            |
| BRCA-mutated Cancer              | USP1      |                       |           |              |             |            | Out-licensed to Exelixis                                 |
| Immuno-Oncology                  | QPCTL     |                       |           |              |             |            |                                                          |
| ER+HER2- Breast Cancer           | KAT6      |                       |           |              |             |            | Out-licensed to Menarini                                 |
| Solid Tumors                     | KIF18A    |                       |           |              |             |            | Out-licensed to Menarini                                 |
| Inflammatory Bowel Disease       | PHD1/2    | Gut-Restricted        |           |              |             |            |                                                          |
| Anemia of Chronic Kidney Disease | PHD1/2    |                       |           |              |             |            | Greater China Rights Out-licensed to TaiGen              |
| MTAP-/- Cancer                   | MAT2A     |                       |           |              |             |            |                                                          |
| Solid Tumors                     | TEAD      |                       |           |              |             |            |                                                          |
| Solid Tumors                     | ENPP1     |                       |           |              |             |            | IND clearance                                            |
| CNS                              | NLRP3     |                       |           |              |             |            | Co-development with Hygia Therapeutics                   |
| Inflammatory Diseases            | NLRP3     | Peripheral restricted |           |              |             |            |                                                          |
| Obesity & Metabolic Diseases     | GLP-1R    |                       |           |              |             |            | Out-licensed to undisclosed partner                      |
| Pain                             | Nav1.8    |                       |           |              |             |            | Greater China rights out-licensed to undisclosed partner |
| Pain                             | Target Z  |                       |           |              |             |            |                                                          |
| Immuno-Oncology                  | CBLB      |                       |           |              |             |            |                                                          |
| Solid Tumors                     | PRMT5     |                       |           |              |             |            |                                                          |
| Obesity & Metabolic Diseases     | GIPIR     |                       |           |              |             |            |                                                          |
| KRAS-driven Solid Tumors         | Pan-KRAS  |                       |           |              |             |            |                                                          |
| Metabolic Disease and Oncology   | NR3C1     |                       |           |              |             |            |                                                          |
| Metabolic Diseases               | Lp(a)     |                       |           |              |             |            |                                                          |
| Inflammatory Diseases            | VAV1      |                       |           |              |             |            |                                                          |
| Obesity & Metabolic Diseases     | APJ       |                       |           |              |             |            |                                                          |

Available for Licensing  
More programs in discovery

**VALUES**

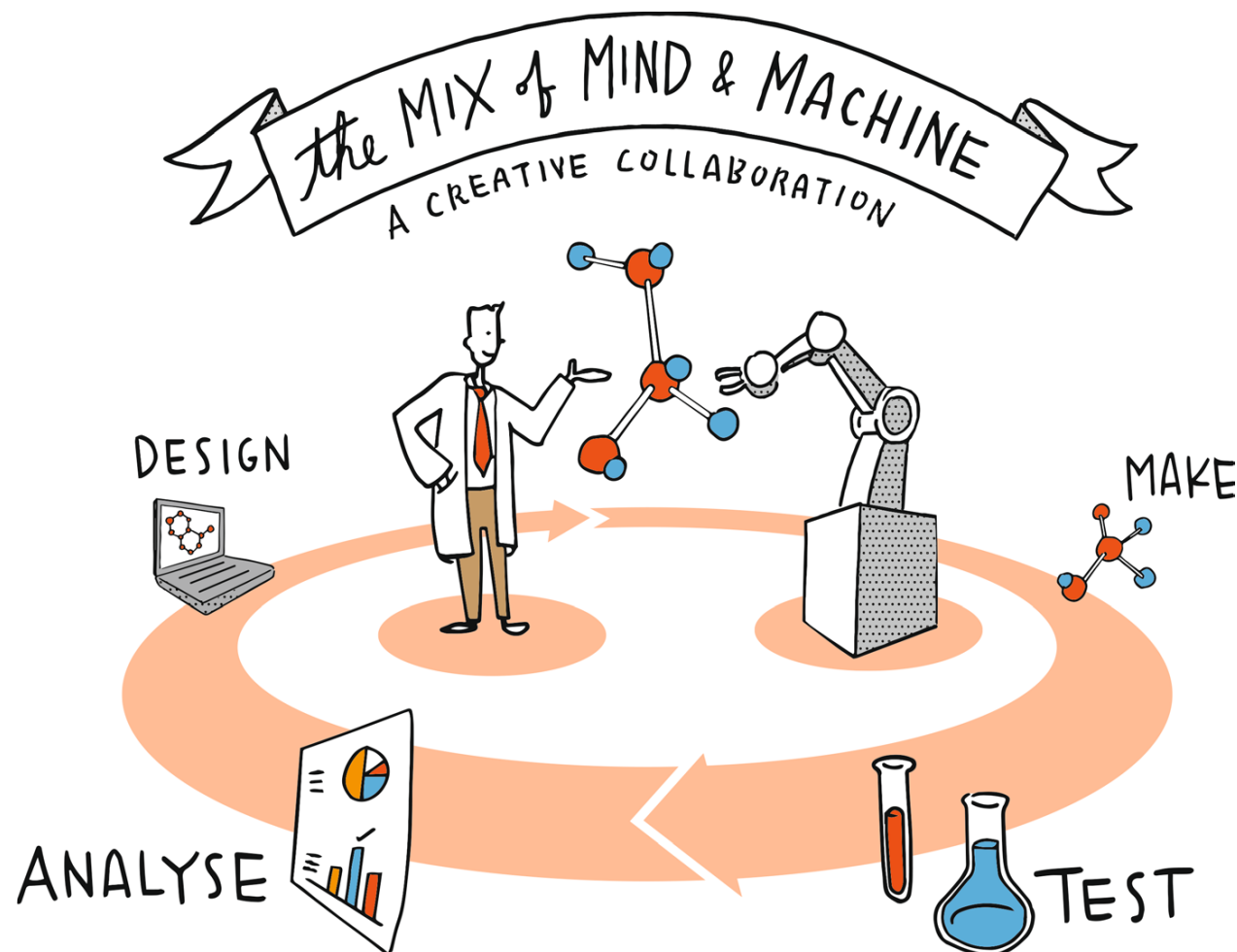
- PATIENT FIRST
- RELENTLESS INNOVATION
- TRANSPARENCY & INTEGRITY
- HIGHEST QUALITY & REPRODUCIBILITY

LinkedIn: Insilico Medicine

Pharma.ai

insilico.com/pipeline

# Reflections



## Rethinking drug design in the artificial intelligence era

Petra Schneider, W. Patrick Walters, Alleyn T. Plowright, Norman Sieroka, Jennifer Listgarten<sup>ID</sup>, Robert A. Goodnow Jr., Jasmin Fisher<sup>ID</sup>, Johanna M. Jansen, José S. Duca, Thomas S. Rush, Matthias Zentgraf, John Edward Hill, Elizabeth Krutoholow, Matthias Kohler, Jeff Blaney, Kimito Funatsu, Chris Luebke and Gisbert Schneider<sup>ID</sup>

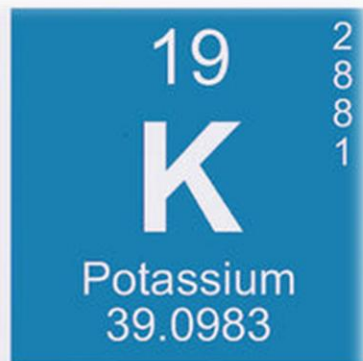
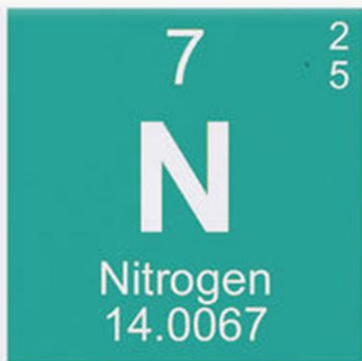
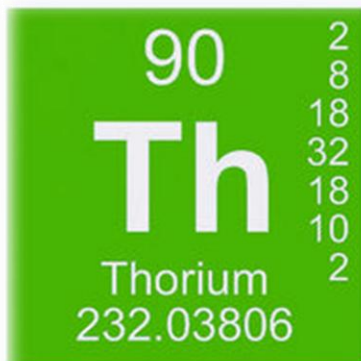
### The Jungle of Generative Drug Discovery: Traps, Treasures, and Ways Out

Rıza Özçelik<sup>1,2</sup> Francesca Grisoni<sup>\*1,2</sup>

## What is the plausibility that all drugs will be designed by computers by the end of the decade?

José L. Medina-Franco & Edgar López-López

- Schneider, P., Walters, W.P., Plowright, A.T. et al. Rethinking drug design in the artificial intelligence era. Nat Rev Drug Discov 19, 353–364 (2020).
- Rıza Özçelik, Francesca Grisoni. The Jungle of Generative Drug Discovery: Traps, Treasures, and Ways Out, arXiv:2501.05457v1 [q-bio.BM] 24 Dec 2024.
- Medina-Franco, J. L., & López-López, E. (2024). What is the plausibility that all drugs will be designed by computers by the end of the decade? Expert Opinion on Drug Discovery, 19(5), 507–510.



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